

Problem Set 2 (three pages)
Statistics 24510-30040 (W25)
Due 9 am Tuesday, January 21.

Requirements

- Put the assignment number, course number (24510 or 30040) and your name at the top of the first page.
- Scan and save your work as a letter-size pdf file named "Lastname-Firstname-hw2.pdf" (abbreviations allowed for long names).
- Upload your pdf to Gradescope under Pset2. Be sure to mark the pages corresponding to each problem during submission.
- Show your work. Provide detailed derivations and arguments in order to receive credit.
- When using R, select only the relevant part of the output to be inserted/displayed.
- You may discuss methods with others, but you should devise and write your answers by yourself completely and independently.

Problem assignments (Related contents in Rice chapters 2-4, 7-9; in Casella and Berger chapter 9.)

1. (*Binomial probability approximations*)

Suppose $X \sim \text{Binomial}(n, p)$. We are interested in the probability $\mathbb{P}(X = k)$ for

Case 1: $k = 3, n = 6, p = 0.3$; Case 2: $k = 11, n = 40, p = 0.4$; Case 3: $k = 2, n = 400, p = 0.0025$;

as listed in the table blow. In each case, compute

- The exact Binomial probability.
- An approximation using the normal distribution for Binomial (with continuity correction).
- An approximation using the Poisson distribution.

(a) Calculate and show the values of the probabilities in a 3-by-3 table (you may use R). Keep four decimal places.

	Binomial probability (exact)	Normal approximation (with continuity correction)	Poisson approximation
$k = 3, n = 6, p = 0.3$			
$k = 11, n = 40, p = 0.4$			
$k = 2, n = 400, p = 0.0025$			

(b) Provide the three formulas of the calculation you used in (a).

(c) (*On approximation accuracy*) Consider the following scenario:

You are asked to approximate the binomial probability (assuming it is too hard to calculate directly for some reason) in each of the two scenario below by either Normal approximation (with continuity correction) or Poisson approximation.

Scenario i: Approximation $\mathbb{P}(X = 10)$ when $X \sim \text{Bin}(30, 1/3)$.

Scenario ii: Approximation $\mathbb{P}(X = 3)$ when $X \sim \text{Bin}(500, 1/1000)$.

Which approximation method would you use? Your reason (in terms of sizes and relations of n, p and k)?

Your answer for this question only need to base on the approximation results in (a), no calculation is needed.

2. (*Wald and Wilson confidence intervals - for Exponential parameter*)

Suppose $X_1, \dots, X_n$ are *i.i.d.* $\sim \text{Exp}(\lambda)$ with probability density function $f(x) = \lambda e^{-\lambda x}, x \geq 0$.

You have derived that $\hat{\lambda} = 1/\bar{X} = n/\sum_{i=1}^n X_i$ is the MLE for λ . From which we've obtained (via the Delta method)

$$\frac{\sqrt{n}}{\lambda} (\hat{\lambda} - \lambda) \sim N(0, 1) \quad \text{asymptotically approximately for large } n.$$

(a) Use the above asymptotic distribution to construct a Wald's $(1 - \alpha)100\%$ confidence interval for λ (by replacing λ in the denominator with $\hat{\lambda}$).

- (b) Use the above asymptotic distribution to construct a Wilson's $(1 - \alpha)100\%$ confidence interval for λ (you may assume $n \geq 4$).

3. (*Variance stabilization method for confidence interval - for Bernoulli-Binomial distribution*)

Let $X_1, \dots, X_n$ be *i.i.d.* $\sim \text{Bernoulli}(p)$, with $\mathbb{P}(X_i = 1) = p, \mathbb{P}(X_i = 0) = 1 - p$.

Then $\sum_{i=1}^n X_i \sim \text{Binom}(n, p)$, and $\hat{p} = \sum_{i=1}^n X_i / n$ is the maximum likelihood estimator (MLE) of p .

Recall that an MLE is asymptotically of normal distribution, and a smooth function of an MLE is also an MLE.

- Derive a variance stabilization transformation for p .
Your resulting transformation should be of the form $(\text{constant}) \times \arcsin \sqrt{\hat{p}}$.
- Let $g(p) = \arcsin \sqrt{p}$. Find the mean and variance of the asymptotic normal distribution of $g(\hat{p})$.
- Use the asymptotic normality of $g(\hat{p})$ to find a $(1 - \alpha)100\%$ confidence interval for $g(p) = \arcsin \sqrt{p}$.
- What should be the range of $g(p)$? Make appropriate adjustment (i.e., thresholding) to the confidence interval of $g(p)$ in (c) to make it consistent with possible values of $g(p)$.
- Use the confidence interval for $g(p)$ to derive a $(1 - \alpha)100\%$ confidence interval for p , the original parameter.

4. (*Coverage probability of binomial asymptotic confidence intervals, R simulation*)

- Generate $k = 100$ draws of *i.i.d.* random variable from $\text{Binom}(n, p)$ for $n = 30, p = 0.1$ (R command `rbinom`).
For each draw, obtaining MLE $\hat{p}$, using test level $\alpha = 0.05$ to compute

- Wald confidence interval for p (with endpoints $\hat{p} \pm z_{\alpha/2} \sqrt{\hat{p}(1 - \hat{p})/n}$)
- Wilson confidence interval for p (with endpoints $(z_{\alpha/2}^2 + 2n\hat{p} \pm z_{\alpha/2} \sqrt{z_{\alpha/2}^2 + 4n\hat{p}(1 - \hat{p})}) / 2(n + z_{\alpha/2}^2)$)
- The arcsine (converted) confidence interval for p you just derived. (R command for arcsine function is `asin`)

In your simulation,

- What is the theoretical coverage probability, that is, the theoretical expected proportion of the k confidence intervals that should contain the true parameter $p = 0.1$?
 - For each type of confidence interval, find the proportion of the k confidence intervals actually contains the true parameter $p = 0.1$, based on your simulation.
 - Compare and comment on the goodness of coverage probability of each of the three confidence intervals, based on your simulation results.
- Repeat the simulation in part (a) with increased sample size $n = 150$.
Do any of your previous conclusions for $n = 30$ change? Why?

Partial sample R code for Q4

```
n = 30
p = 0.1
z = 1.96 # or = qnorm(.975)
runs = 100 # number of draws of binomial observations

obs=rbinom(runs,n,p)
pHat=obs/n # observed binomial proportions
pHat=sort(pHat)
propWald=rep(0,runs)

# limits of Wald C.I.
Lwald = pHat - z*sqrt(pHat*(1-pHat)/n)
Uwald = pHat + z*sqrt(pHat*(1-pHat)/n)

for (i in 1:runs)
{
  propWald[i] = ifelse(p < Uwald[i] && p > Lwald[i],1,0)
}
sum(propWald)/runs
```

5. (*Sum and ratio of random variables*)

Suppose X and Y are independent continuous random variables.

- (a) Derive the bivariate probability density function of (S, R) , with $S = X + Y$ and $R = \frac{X}{X + Y}$.
- (b) Furthermore, suppose $X \sim \text{Gamma}(\alpha_1, \lambda)$ and $Y \sim \text{Gamma}(\alpha_2, \lambda)$ are independent Gamma random variables. A $\text{Gamma}(\alpha, \lambda)$ random variable has probability density function $f(x) = \lambda^\alpha x^{\alpha-1} e^{-\lambda x} / \Gamma(\alpha)$, $x \geq 0$.
 - i. Find the bivariate density functions of $S = X + Y$ and $R = \frac{X}{X + Y}$, based on part (a).
 - ii. From the formula in part (i) above, find the density function of $S = X + Y$.
What is the name (and parameters) of the probability distribution of S ?
 - iii. From the formula in part (i), find the density function of $R = \frac{X}{X + Y}$.
What is the name (and parameters) of the probability distribution of R ?
 - iv. What is the name of the probability distribution in part (iii) above when $\alpha_1 = \alpha_2 = 1$?